

MUHAMMAD AZAMUDDIN BIN MOHD AZMI

Senior Software Engineer & AI Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience building scalable full-stack platforms, machine learning systems, and AI-powered applications. Expert in distributed systems, microservices architecture, and cloud-native development on AWS, GCP, and Azure. Proven track record designing high-performance REST and GraphQL APIs, production ML pipelines, realtime data systems, and LLM-based solutions using RAG, vector embeddings, and fine-tuning. Strong background in Python, Go, JavaScript/TypeScript, React, Node.js, Docker, Kubernetes, and CI/CD automation.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

SOMA AI Technologies Inc. | San Francisco, California, US

May 2025 – Mar 2026

- Architected and developed scalable AI infrastructure for decentralized intelligence networks on the Bittensor ecosystem, enabling distributed inference, semantic search, and high-throughput validation across subnet participants.
- Designed and implemented distributed backend microservices using Python, FastAPI, Node.js, PostgreSQL, Redis, and cloud-native infrastructure to support validator, miner, and inference APIs with high availability.
- Built end-to-end machine learning pipelines covering large-scale data collection, preprocessing, feature engineering, model training, evaluation, and deployment using PyTorch, Hugging Face Transformers.
- Developed embedding-based semantic intelligence systems using transformer models, vector databases, and cosine similarity search to improve contextual understanding, ranking quality, and decentralized knowledge retrieval.
- Fine-tuned transformer-based language models including BERT, RoBERTa, and open-source LLMs on millions of labeled samples, significantly improving multilingual classification accuracy and benchmark performance.
- Engineered real-time inference infrastructure with asynchronous processing, request batching, and optimized model serving, reducing latency by over 40% while supporting large-scale distributed AI workloads.
- Designed Retrieval-Augmented Generation (RAG) pipelines incorporating text chunking, tokenization, embedding generation, and vector retrieval for efficient long-context reasoning and knowledge-intensive applications.
- Integrated vector databases such as Pinecone and Weaviate to provide scalable semantic indexing and similarity search across millions of documents and decentralized network queries.
- Built automated evaluation and benchmarking frameworks to measure model quality, validator scoring consistency, adversarial robustness, and performance across rapidly evolving foundation models.
- Implemented production-grade CI/CD pipelines using Docker, Kubernetes, Helm, GitHub Actions, and AWS services, streamlining deployment workflows and reducing release cycles through automated testing and infrastructure orchestration.

Senior Software Engineer & AI Engineer

b2wise Company | Birmingham B4 6AF, United Kingdom

Feb 2020 – Jan 2025

- Designed and implemented scalable backend services using Node.js and Python.
- Built REST APIs and database architectures with PostgreSQL and MongoDB.
- Developed AI-powered chatbot and document-processing solutions using LLM APIs and LangChain.
- Integrated authentication, payment gateways, and real-time communication features.
- Deployed applications using Docker and cloud infrastructure with CI/CD pipelines.
- Collaborated with international teams to deliver enterprise SaaS solutions.

Frontend Developer & User Management

Adin Company | Oslo, Norway

March 2019 – Jan 2020

- Developed responsive healthcare management platforms using React, TypeScript, and Next.js.
- Built user management modules for clinics and hospitals with role-based access control.
- Integrated REST APIs and authentication systems for secure patient and staff workflows.
- Collaborated with backend engineers and designers to deliver production-ready features.
- Improved application performance and user experience across multiple enterprise projects.

EDUCATION

Bachelor's Degree in Computer Science / Software Engineering

Universiti Sains Malaysia

Graduated:2018

SKILLS

- Programming Languages: Python, JavaScript, TypeScript, Java, Go, C, C#, Rust, PHP, Kotlin, Dart,
- Solidity Frontend: React, Next.js, Angular, Vue.js, Nuxt.js, Redux, Tailwind CSS, Bootstrap, Flutter
- Backend: Node.js, Go, Express.js, NestJS, FastAPI, Django, Flask, Spring Boot, Ruby on Rails, REST APIs, GraphQL, WebSockets, Microservices
- Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL Server, SQLite, Firebird, Pinecone, Weaviate
- ML / AI: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, OpenCV, YOLO, RAG, Fine-Tuning, Vector Embeddings, LLMs (GPT-4, Claude, Gemini), NumPy, Pandas
- Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, ECS, SageMaker), GCP (Compute Engine, Vertex AI), Azure (App Services, Functions), Docker, Kubernetes, Helm, GitHub Actions, Jenkins, GitLab CI/CD, CircleCI Security: JWT, OAuth 2.0, AES Encryption, SHA-256, Merkle Trees, OWASP best practices
- Testing: Jest, Mocha, Chai, Cypress, Playwright, Selenium, TDD, Unit/Integration/E2E

CERTIFICATIONS

- AWS Certified Cloud Practitioner – Amazon Web Services (AWS)
- IBM AI Engineering Professional Certificate – IBM
- Certified by micro1 – micro1